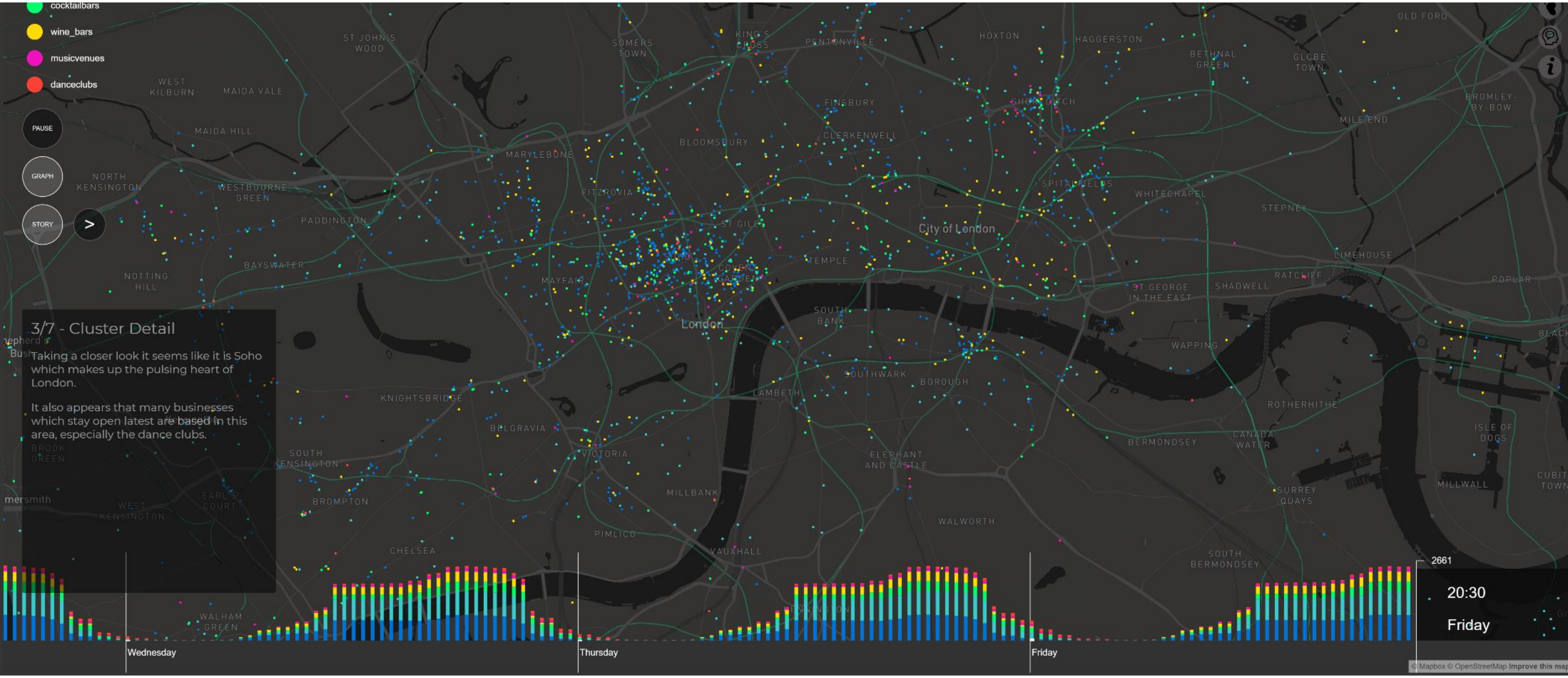
CASA0029: Urban Data Visualisation

Online Data Sources and APIs



Yelp API data visualisation example (Kahina Ait Ouazzou, Shan Yi See, Greg Slater, Dan Qin)
https://london-rhythms.github.io/

Duncan A Smith

Centre for Advanced Spatial Analysis, UCL

Overview

1. **Data Strategies for Online Visualisation**
2. **Common Data Formats**
3. **Web API Examples**

Practical-

4. Using Web APIs

Data Strategies for Online Visualisation

Flat Files

Online Databases / Cloud Services

Application Programming Interfaces (APIs)

Flat Files

Simple, common approach- store data as static CSV, JSON and GeoJSON files in same folder as HTML pages on web server.

Quick, easy, will work on files up to several MB which can be more data than you expect.

Limitations-

Not scalable for large files (especially large spatial data)

Static. Only works for archived data, not for dynamic / real time data

[World City Population]map uses flat GeoJSON & CSV files as relatively small point dataset ->

Online Databases and Cloud Services

For storing large files online in a scalable way (or for managing dynamic data), typically need online databases. Either self-hosting (e.g. MySQL, PostGIS) on web server, or using hosting capabilities of a cloud service (e.g. Mapbox, Carto).

Self Hosting: MySQL, PostGIS
Database software is not taught on this module (see Big Data/Data Science modules). Self-hosting is an option if you want to develop your skills in database management and servers.

Cloud Hosting: Mapbox, Carto, MapTiler
Cloud services will effectively host a database for you. Mapbox will create vector tiles for large spatial datasets. Carto commercial platform integrated with database/Big Data storage.

Mapbox: Vector Tiles Solution

Mapbox will host your large vector spatial datasets as Tilesets, and serve these layers for interactive maps.

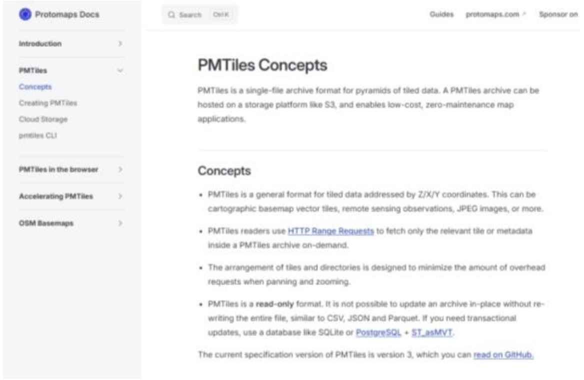
Scalable solution for large spatial datasets. Some functions to filter large datasets, and to query spatial features-
https://docs.mapbox.com/api/maps/#tilesets
https://docs.mapbox.com/playground/tilequery/

Mapbox more limited in terms of attribute queries, and for hosting dynamic data.

MapLibre and Open Vector Tiles

When Mapbox became more commercial, MapLibre was formed as fully open source alternative using largely the same codebase- https://maplibre.org/

MapLibre supports alternative vector tiles format PMTiles, and are currently developing MapLibreTiles. These are hybrid formats with database performance in a flat file. Interesting free solution, no server needed-
https://docs.protomaps.com/pmtiles/
https://maplibre.org/news/2026-01-23-mlt-release/


Protomaps Docs

Search Ctrl K | Guides | protomaps.com | Sponsor on

Introduction

PMTiles
- Concepts
- Creating PMTiles
- Cloud Storage
- pmtiles CLI

PMTiles in the browser

Accelerating PMTiles

OSM Basemaps

PMTiles Concepts

PMTiles is a single-file archive format for pyramids of tiled data. A PMTiles archive can be hosted on a storage platform like S3, and enables low-cost, zero-maintenance map applications.

Concepts

- PMTiles is a general format for tiled data addressed by Z/X/Y coordinates. This can be cartographic basemap vector tiles, remote sensing observations, JPEG images, or more.
- PMTiles readers use HTTP Range Requests to fetch only the relevant tile or metadata inside a PMTiles archive on-demand.
- The arrangement of tiles and directories is designed to minimize the amount of overhead requests when panning and zooming.
- PMTiles is a **read-only** format. It is not possible to update an archive in-place without re-writing the entire file, similar to CSV, JSON and Parquet. If you need transactional updates, use a database like SQLite or PostgreSQL + ST_asMVT.

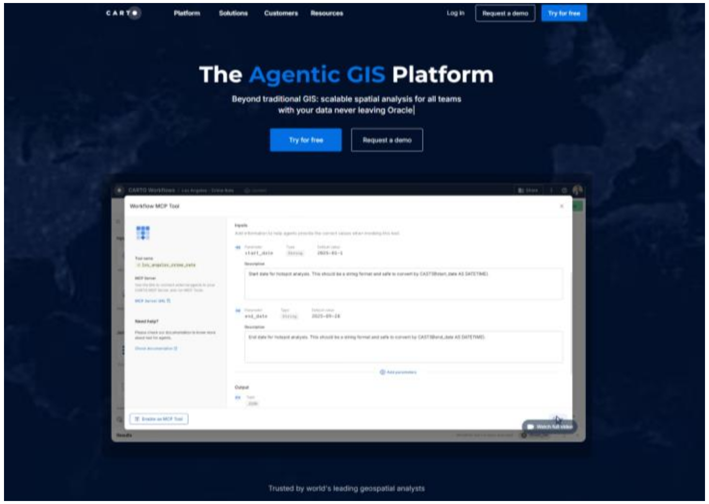
The current specification version of PMTiles is version 3, which you can read on GitHub.

Carto Hosting and Databases

Carto similar company to Mapbox (not free but has student option). Has some powerful analytics capabilities-

https://api-docs.carto.com/#intro

New version of Carto has links with deck.gl, Google BigQuery- https://docs.carto.com/



Web APIs

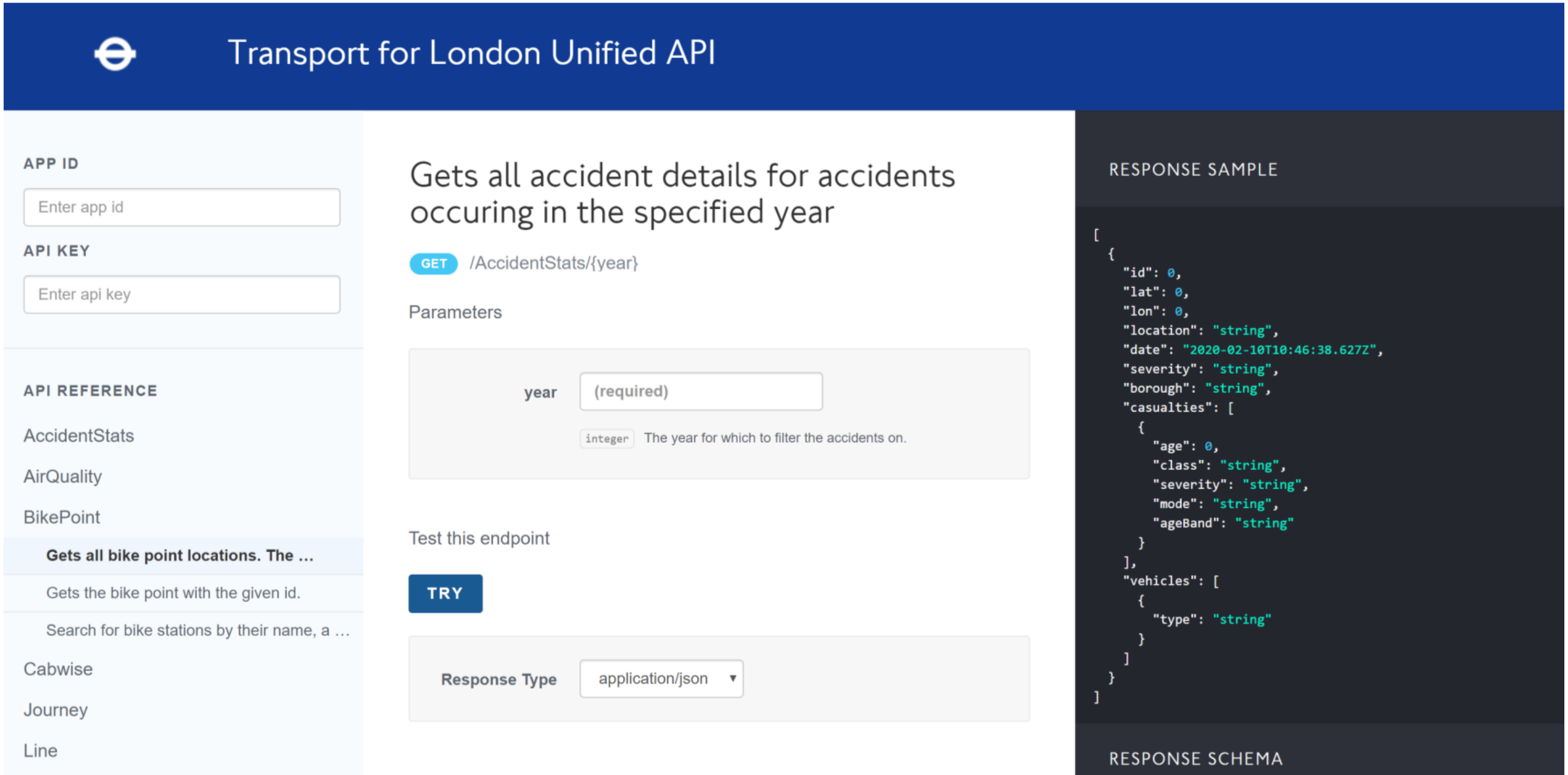
Application Programming Interfaces (APIs)- interface or communication protocol for accessing data and/or functions.

Web APIs are a standard means of accessing datasets online from large public data holders (e.g. NASA, UK government, EU…), and for interacting with web services. We have already been using the Mapbox API to access the Tilesets and Styles we uploaded to Mapbox.

Real time (or near real time) data typically accessed through Web APIs. Ideal for time-sensitive data: transportation, weather, air quality, stock markets...

API Example- Transport for London

TfL host a popular series of APIs for real-time data on London public transport networks. Includes tube line status, journey-planner results, travel times, bike share docking station feeds…
https://api.tfl.gov.uk/



Web API Standards

Web APIs describe public endpoints to web services, typically using HTTP standards to request data through a URL, and return data as XML or JSON. Most common HTTP methods are called GET (to request data) and POST to send data.

Common for the requested query to be encoded in the API URL (using the format described in the API documentation).
Examples from Tfl APIs https://api.tfl.gov.uk/-

Tell me current status of the Northern Line-
https://api.tfl.gov.uk/Line/northern

Tell me Number of Bikes and Empty Docks at St. James bike point-
https://api.tfl.gov.uk/BikePoint/Search?query=St.%20James

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Parsing Data from APIs

Typically data from APIs returned in JSON or XML format. D3 library (used in practical last week) offers useful methods for importing these datasets. Simplifies process of making HTTP requests and parsing the data. Other JavaScript libraries will do similar job (e.g. JQuery).

```
d3.csv()
d3.json()
d3.xml()
```

Once imported, typically have to scan through the data object to find the particular variables you are interested in. Useful to begin by logging the data object in the console so you can see where relevant items are in the array-

console.log(*mydata*);

JSON

JavaScript Object Notation (JSON)- open-standard data interchange format to transmit data objects consisting of attribute-value pairs and array data types. Very common web data format, used by many APIs.

JSON data defined as name-value pairs within curly brackets. Can also include nested JSON objects and arrays (in square brackets)-

```
{
 "firstName": "Mary",
 "lastName": "Jones",
 "age": 35,
 "address": {
     "city": "New York",
     "postal code": "10021"
     }
}
```

https://www.w3schools.com/js/js_json_intro.asp

Getting Data from a JSON Object

In JavaScript, refer to the name to return the value-

```
var myJSON = { “name”: “Mary”}
console.log(myJSON.name); // Returns Mary
console.log(myJSON[“name”]); // Returns Mary
```

Often APIs will return an array of JSON objects. In this case you can refer to array position, e.g-

```
myJSON[5].name;
```

And often you need to loop through the array-

```
for (i = 0; i < myJSON.length; i++) { ...}
```

GeoJSON

GeoJSON is a standard for storing simple vector spatial data and attributes in JSON format.

GeoJSON can be used by some APIs and web database queries. Features include points, linestrings and polygons. Multiple features stored as an array known as a FeatureCollection.

```
var myGeoJSON = {
    "type": "Feature",
    "geometry": {
      "type": "Point",
      "coordinates": [125.6, 10.1]  // Longitude then latitude
    },
    "properties": {
      "name": “Manchester"
    }
  }
```

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Web API Examples

Not required to use APIs for your Group Projects. It depends whether real time data is relevant for your topic. Most spatial data is archived data rather than real time data, such as the data sources we have been using (e.g. Digimap, London Datastore, data.gov.uk, OpenStreetMap, Land Registry…).

There may be useful datasets that you want to use that need to be accessed via APIs.

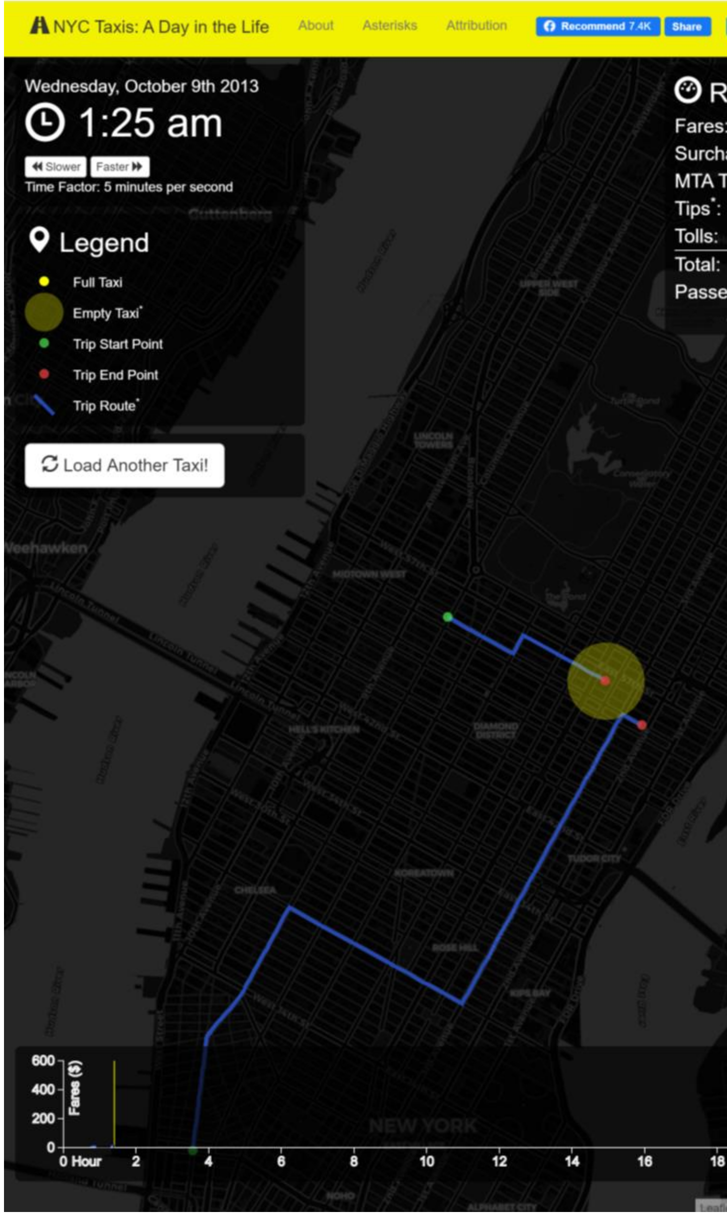
Typical examples- transport data, social media data, business data, meteorological/environmental data…

Transport APIs

International Flight Tracker Data-
https://aviation-edge.com/flight-radar-and-tracker-api/

UK Transport Data-
Transit Timetable:
https://www.travelinedata.org.uk/
Real Time Trains:
https://api.rtt.io/
Live Traffic:
https://data.gov.uk/search?q=Highways+England

NYC Taxi Data-
https://data.cityofnewyork.us/Transportation/2021-Yellow-Taxi-Trip-Data/m6nq-qud6



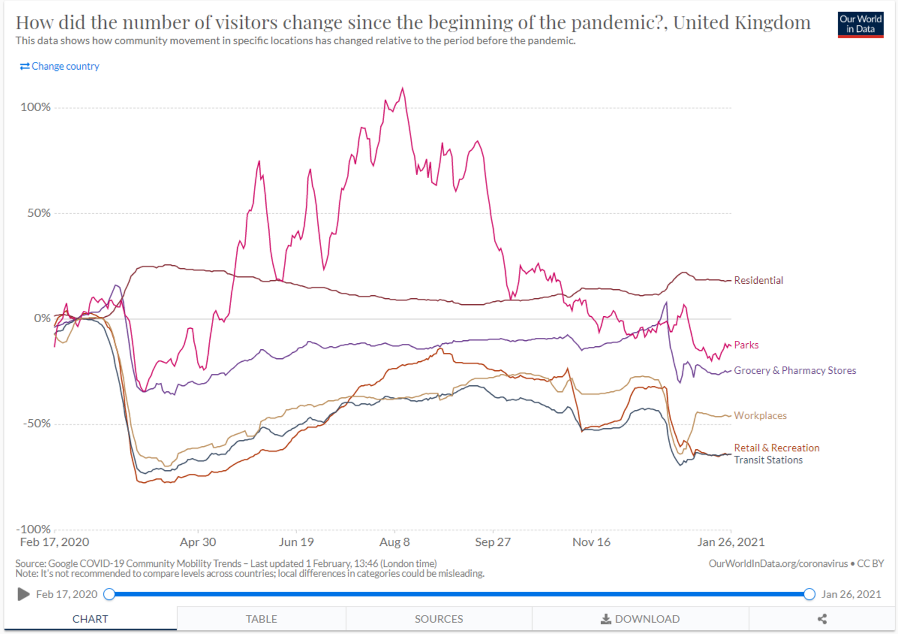
https://chriswhong.github.io/nyctaxi/

Transport mode Activity Levels

The Covid-19 lockdowns in 2020 brought a huge fall in public transport passenger numbers. There are several datasets tracking changing transport activity levels in recent years.

TfL aggregate passenger numbers- https://data.london.gov.uk/dataset/public-transport-journeys-type-transport

Department for Transport aggregate transport use- https://www.gov.uk/government/statistics/transport-use-during-the-coronavirus-covid-19-pandemic



https://ourworldindata.org/covid-mobility-trends

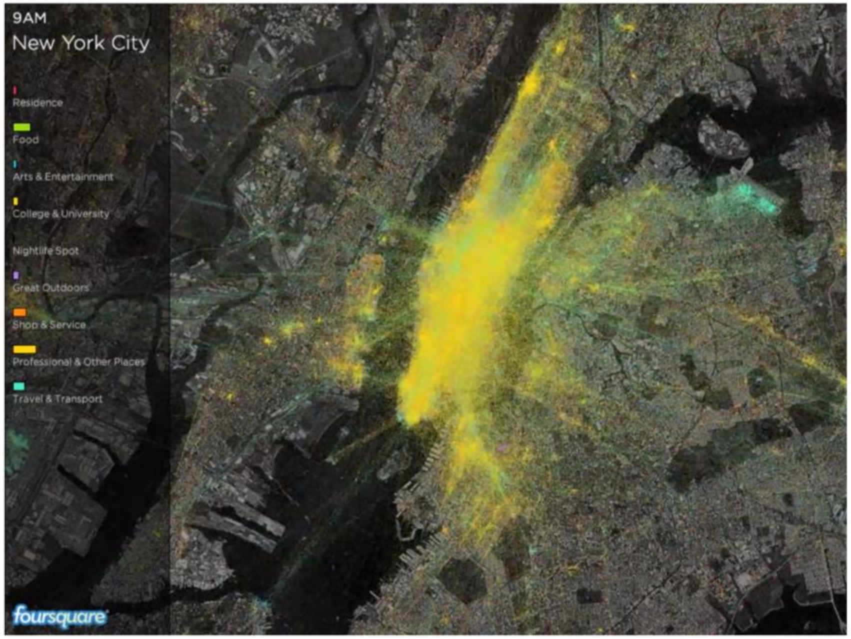
Social Media APIs

Social media data become more restricted due to privacy and monetization of data. Still some popular examples-

Flickr Data-
https://www.flickr.com/services/developer/api/

FourSquare Data-
https://developer.foursquare.com/docs

Twitter/X Data (free version now removed)-
https://docs.x.com/home



https://foursquare.com/infographics/pulse

Business and Property APIs

AirBnB Listings and Price Data-
http://insideairbnb.com/get-the-data.html

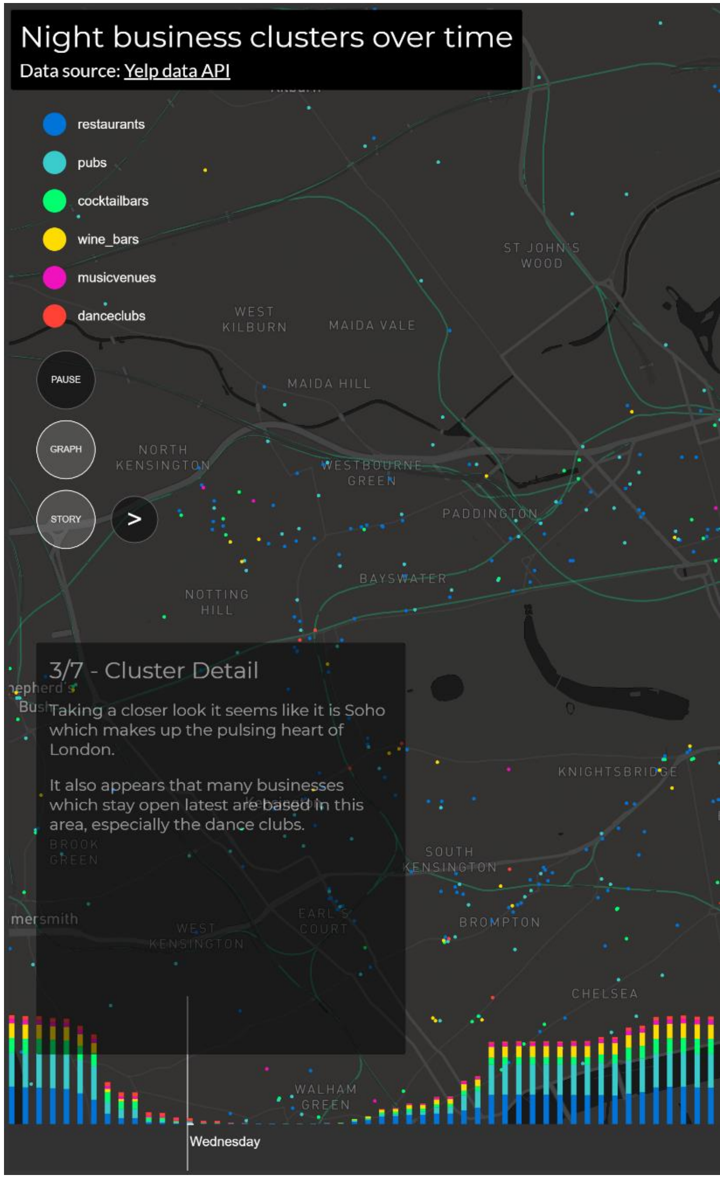
Yelp Business Type and Opening Hours Data-
https://www.yelp.com/dataset

Many property datasets available in static CSV formats-

London Development Database-
https://www.london.gov.uk/programmes-strategies/planning/digital-planning/planning-london-datahub

Valuation Office Data (commercial floorspace and rents)-
https://voaratinglists.blob.core.windows.net/html/rlidata.htm

Land Registry House Price Data-
https://landregistry.data.gov.uk/



https://london-rhythms.github.io/

Environmental APIs

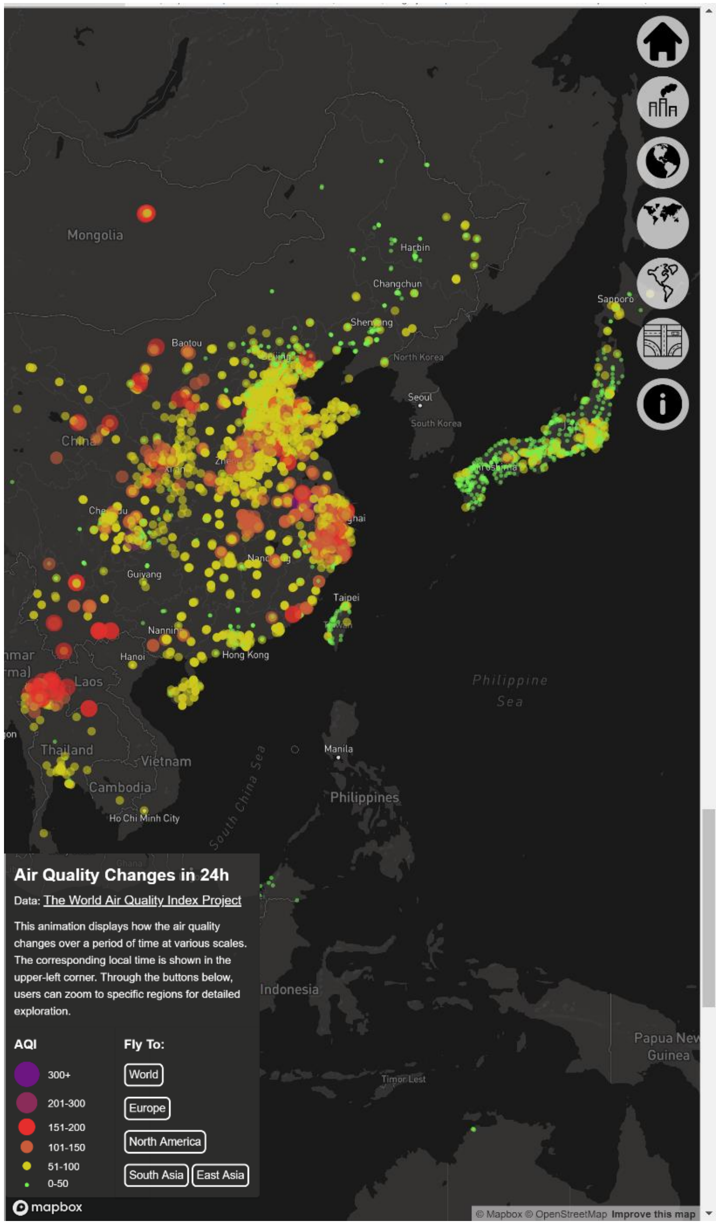
International Air Quality data-
https://aqicn.org/api/

London Air Quality Network-
https://www.londonair.org.uk/LondonAir/API/

UK Air Quality-
https://uk-air.defra.gov.uk/data/

Global Weather-
https://openweathermap.org/api

Copernicus Satellite and Air Quality data-
https://dataspace.copernicus.eu/



https://naughty-kepler-88ce41.netlify.com/

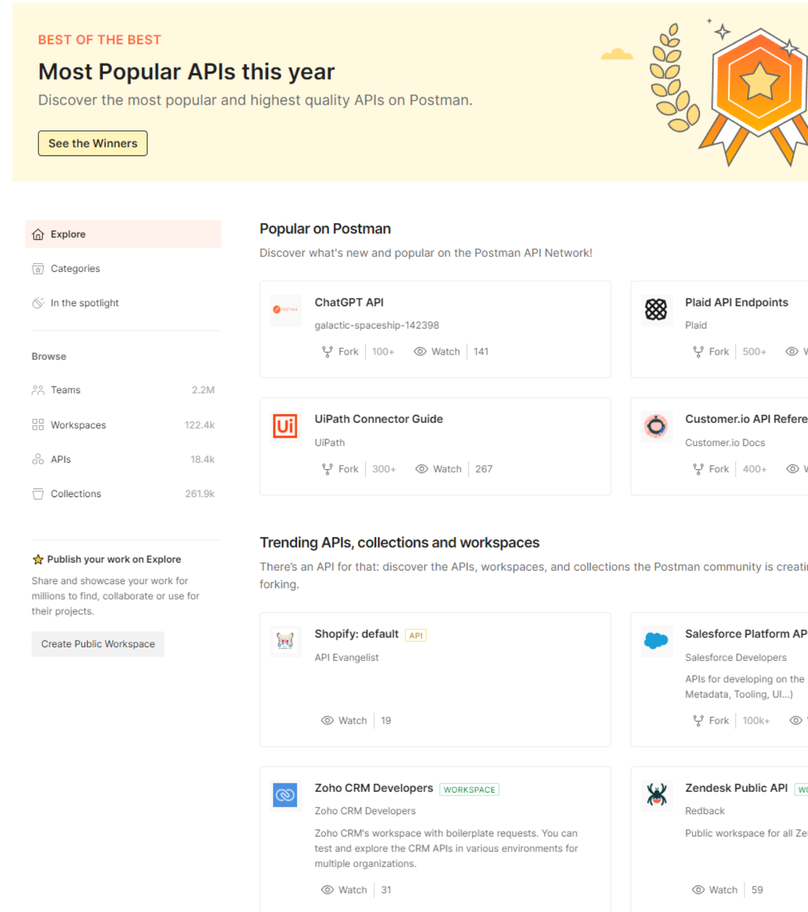
API Directory Websites

There are some useful websites that catalogue a great many APIs, and can be a good place to search for particular datasets-

https://explore.postman.com/

https://any-api.com/

https://github.com/APIs-guru/openapi-directory



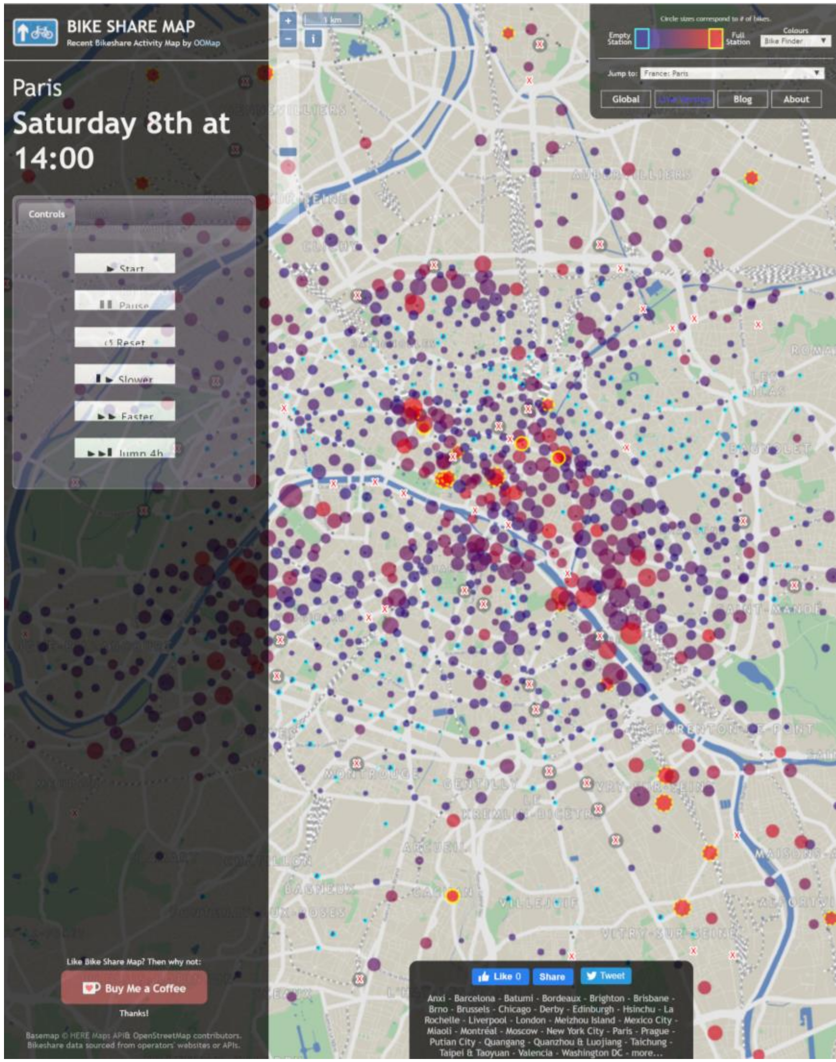
Archiving Real Time Data Feeds

Real time data is often more revealing when put in context of historical data. Limited number of APIs offer real time and archived data.

Can be good strategy to collect your own data archive by writing a script (e.g. Python) to collect API data over time.

UCL researcher examples of archiving data-

Ollie O'Brien, Bikeshare
Richard Milton, TfL Tube Data



https://bikesharemap.com/

Conclusions

Data Strategies: Flat Files, Databases and APIs
Consider the best method for storing and accessing data for your visualisation. Depends on data size, currency and nature of data queries.

Web APIs for Data Requests and Real Time Data
Flat files easiest, but APIs needed to query large datasets and access real time data.

Common Types of APIs
Data APIs used for live data: transport, social media, environment... We also use APIs to interact with cloud services such as Mapbox.

Can Use JavaScript to Parse API Data in JSON format
API data typically in JSON format, easy to manipulate in JavaScript (also in Python/R)

Group Discussion – Live Data

We are going to look at a website mapping global air quality data, the World Air Quality Index Project https://aqicn.org/map/world/

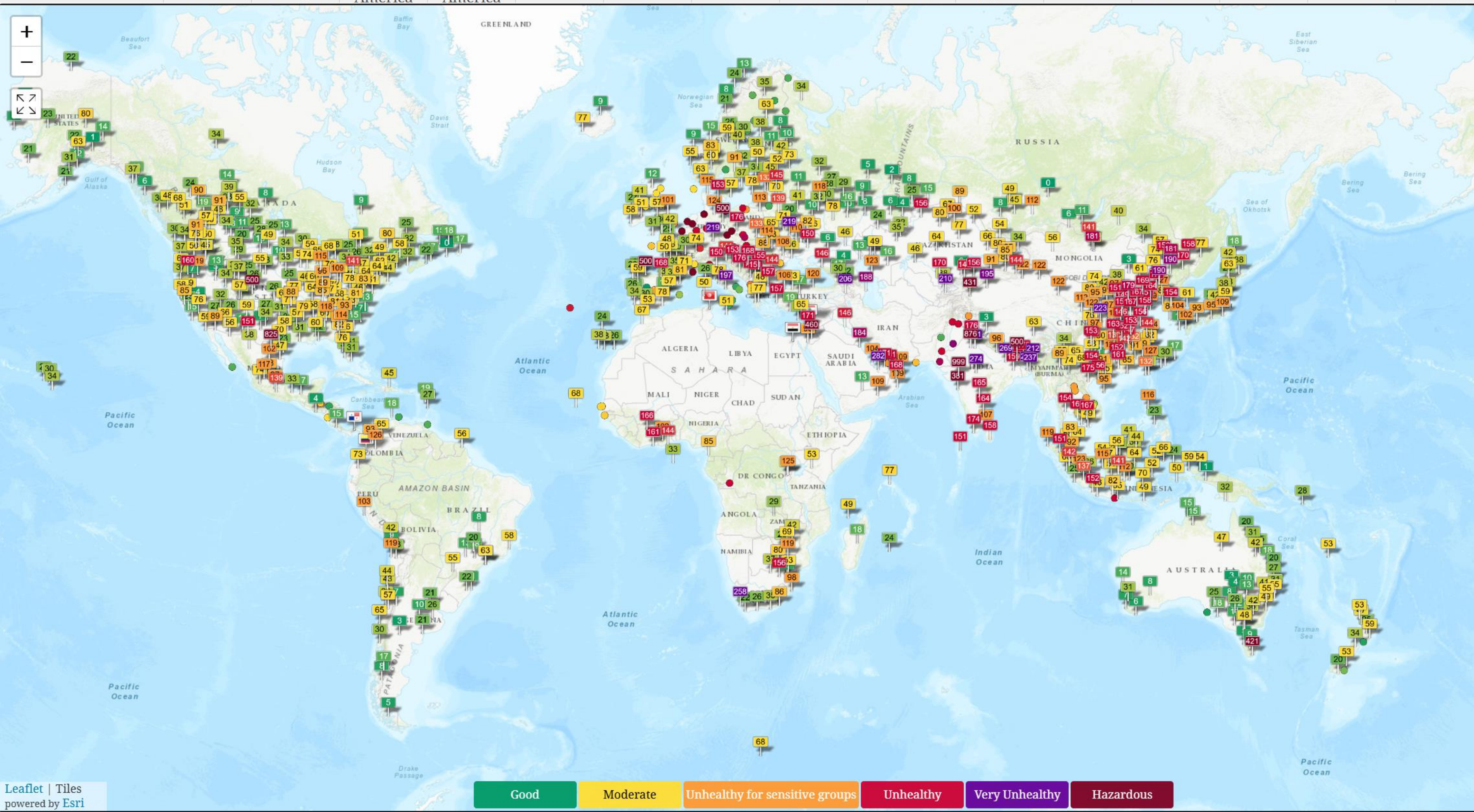
Explore this site which, provides a global overview of live air quality data, and more detailed information on a city by city basis. **What are the advantages of having real time data for this application?**

This website brings together hundreds of different data feeds on air quality. Can you think of any **challenges created through having such a large number of different datasets?** How has this website tackled these challenges?

I discussed that comparing real time data to archived data can be useful. Explore this feature for individual cities in the 'Air Quality Historical' section- https://aqicn.org/historical/#!city:london
Explore their API service- https://aqicn.org/api/

Air Pollution in World: Real-time Air Quality Index Visual Map

Asia | Europe | North America | South America | Africa | Australia | Middle East | India | China | Japan | South Korea | Thailand | Mexico | Poland | Turkey



Over Reading Week, Start Thinking About Group Projects

In the second half of the module we will be working on your Group Projects. At the moment you can develop early ideas for project concepts, and form groups of students with similar ideas.

Overall theme – Urban Analytics

Groups should ideally have three students each.

Web APIs Practical